

Prajwal Sah

Student Ambassador @GeeksForGeeks | AI & CyberSecurity | Full-Stack Web Developer

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Experience

Software Developer

Jan 2026 – May 2026

Bhaktapur Municipality (Collaboration through Khwopa Engineering College, Bhaktapur, Nepal)

[Live Website](#)

- Engineered and officially deployed a full-stack tourism and digital heritage platform for Bhaktapur Municipality, integrating **15+ tourism service modules** to digitally promote heritage sites, local businesses, and cultural attractions.
- Designed and developed **15+ responsive web pages** using Django, HTML, CSS, JavaScript, Bootstrap, and PostgreSQL, delivering a seamless user experience across desktop and mobile devices.
- Collaborated within a **4-member development team** throughout the complete software development lifecycle, including requirement analysis, implementation, testing, deployment, and stakeholder feedback.
- Implemented secure user authentication and location-aware tourism features by integrating interactive maps, geolocation services, and nearby attraction discovery based on configurable search radii.
- Contributed to backend integration, database design, and deployment of a production-ready platform officially adopted and publicly hosted by Bhaktapur Municipality.

Technical Expertise

Languages	Python, C, C++, JavaScript, HTML5, CSS3
Data Structures & Algorithms	Arrays, Linked Lists, Trees, Graphs, Dynamic Programming, Problem Solving
Frameworks & Libraries	Django, Flask, Bootstrap, Tailwind CSS
AI & Machine Learning	LLM APIs, Retrieval-Augmented Generation (RAG), Agentic AI, NLP, Prompt Engineering
Databases	PostgreSQL, MySQL, MongoDB
Cybersecurity Tools	Kali Linux, Wazuh SIEM, MITRE ATT&CK, Wireshark
Developer Tools	Git, GitHub, VS Code, $\LaTeX$, WordPress

Projects

Fallacy Detector & AI Reasoning Assistant

KU Hackathon 2025 AI/ML Track Winner

Python | ELECTRA | Flask | OpenRouter API | Manifest V3



- Built an AI-powered logical fallacy detection system achieving **82% classification accuracy** on **1,000+ annotated arguments**, supporting detection of **13 logical fallacy categories**.
- Developed a **Manifest V3 browser extension** capable of analyzing both selected webpage text and user-written content inside editable fields, enabling real-time reasoning assistance.
- Integrated the **ELECTRA transformer model**, Natural Language Processing, and **OpenRouter LLM APIs** to classify logical fallacies and generate contextual explanations through structured argument analysis.
- Collaborated within a **4-member team** to architect, develop, and present the solution during a **48-hour intense hackathon**, winning the **AI/ML Track**.

Spotify Clone Web Application

Personal Project

Django | HTML | CSS | JavaScript



- Developed a full-stack music streaming web application featuring secure authentication, playlist CRUD operations, audio playback, and a Spotify-inspired user interface.
- Designed and implemented **10 interconnected web pages** with Django backend integration, delivering seamless music management and playback functionality.

Repository Management System

Personal Project

Python | PostgreSQL



- Engineered a Python-based repository management system implementing complete CRUD operations with PostgreSQL integration.
- Designed structured repository organization and persistent file-state management using relational database architecture and modular software design principles.

Canteen Management System

Team Project

C++ | *graphics.h*

- Collaborated within a **4-member team** to develop a GUI-based desktop application automating canteen operations including secure login, billing, inventory management, receipt generation, search functionality, and administrative controls.
- Designed an interactive graphical interface using **graphics.h**, improving operational efficiency through streamlined order processing and inventory tracking.

Jarvis AI Personal Assistant

Personal Project

Python | Speech Recognition | *pyttsx3* | REST APIs

- Developed a Python-based voice-activated AI assistant capable of executing natural language commands using speech recognition and NLP.
- Implemented features including web browsing automation, real-time weather updates, Wikipedia queries, system controls, and multi-API integrations.

Additional Mini Projects: Snake Game | Tic-Tac-Toe | Restaurant Booking Website (WordPress) | Portfolio Website

Education

Bachelor of Engineering in Computer Engineering

2022 – Present

Khwopa Engineering College, Bhaktapur

Higher Secondary Education (+2 Science)

GPA: 3.50

2022

Khwopa Higher Secondary School, Bhaktapur

Secondary Education Examination (SEE)

GPA: 3.80

2020

Sainik Awashiya Mahavidyalaya, Bhaktapur

Research & Ongoing Work

- Developing LSA-Agent, a privacy-preserving Agentic RAG framework for automated security incident investigation using Wazuh SIEM and local LLMs.
- Researching AI-assisted threat hunting, MITRE ATT&CK mapping, and on-premise cybersecurity analytics for privacy-preserving environments.
- Evaluating the effectiveness of local large language models for cybersecurity incident analysis and alert triage.

Achievements & Certifications

- **Winner – AI/ML Track, KU Hackathon 2025** [[Certificate](#)]
- **Student Ambassador – GeeksForGeeks Campus Mantri 2026** [[Certificate](#)]
- **2nd Runner-Up – Hult Prize at KhEC 2023** [[Certificate](#)]
- **National CEEL Conference 2026** [[Certificate](#)]
- **LaTeX Typesetting Training** [[Certificate](#)]
- **Arduino Programming & Prototyping Training**

Leadership & Extracurricular

- Experienced in Agile-style team-based software development with responsibilities spanning UI/UX, backend integration, and project coordination.
- Actively participated in hackathons, ideathons, and competitive technical challenges, demonstrating collaborative problem-solving under real-world constraints.
- Graphic Designer, IT Circle – Khwopa Engineering College: Designed promotional materials and visual assets for college technical events and club activities.